SNR Stability Analysis of Energy VAD vs. Lightweight CNN VAD

CS 6140 — Voice Activity Detection Project

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Contents

1 Research Question **2**

2 Methods **2**
2.1 Energy VAD 2
2.2 CNN VAD 2

3 Experimental Setup **2**
3.1 Evaluation audio 2
3.2 SNR sweep 3
3.3 Metrics 3

4 Results **3**
4.1 Single-recording baseline (clean audio) 3
4.2 SNR stability sweep 4
4.3 Summary statistics 4
4.4 Inference latency 5

5 Analysis: Answering the Research Question **5**
5.1 Is CNN VAD more stable? 5
5.2 Does CNN VAD offer better peak performance? 5
5.3 Crossover point 5

6 Real-World Trade-off **6**
6.1 Decision matrix 6
6.2 Cascade design (future work) 6

7 Conclusion **6**

1 Research Question

The original project proposal asked:

> *Under real-time streaming constraints, does a lightweight CNN-based VAD offer meaningfully better stability than a classical energy-based VAD across varying noise conditions — and at what practical cost?*

Stability was defined through two primary metrics: F1 score variance across signal-to-noise ratio (SNR) levels, and worst-case F1 at the hardest noise condition. This report presents the full experimental evaluation addressing that question.

2 Methods

2.1 Energy VAD

The energy-based baseline computes log frame energy over 25 ms windows with 10 ms hop:

$$E_t = \log\left(\sum_n x_t[n]^2 + \varepsilon\right), \quad \varepsilon = 10^{-10}$$

A frame is classified as speech if $E_t > \bar{E}$ (adaptive per-recording mean threshold). A length-5 median filter is applied post-hoc to suppress isolated spurious frames.

Cost: No training required. Runtime: ≈0.005 ms per frame (numpy only).

2.2 CNN VAD

The CNN VAD uses a `LightweightCNN_VAD` model (<500 K parameters, 51,969 actual) operating on 200-dimensional log-mel features: 40 mel bins × 5-frame context window, extracted via a unified numpy FFT pipeline (Hann window, 512-point FFT).

Training data (LibriSpeech train-clean-5 + MUSAN):

- 25% speech only
- 25% MUSAN noise only
- 25% speech + noise mix (SNR 5–20 dB)
- 25% real silence — low-energy pause segments mined from LibriSpeech recordings (RMS < 0.005), critical for eliminating FAR = 0.96

Training uses `BCEWithLogitsLoss` (no sigmoid in model output layer). A global z-score scaler is fit on the training split and saved alongside the checkpoint for inference-time normalisation.

Cost: Requires labelled data and GPU training. Runtime: ≈0.34 ms per frame (PyTorch CPU).

3 Experimental Setup

3.1 Evaluation audio

The test recording is `data/sample.wav`: a 6.21 s, 16 kHz mono recording with a single speech segment from 2.6 s to 4.15 s. Ground-truth labels are frame-aligned at 10 ms hop.

3.2 SNR sweep

MUSAN noise (free-sound category) is mixed into the clean speech at five SNR levels: $\{0, 5, 10, 15, 20\}$ dB. The mixing procedure:

$$y = x_{\text{speech}} + \alpha \cdot x_{\text{noise}}, \quad \alpha = \sqrt{\frac{P_{\text{speech}}}{P_{\text{noise}} \cdot 10^{\text{SNR}/10}}}$$

Ground-truth labels are identical across all SNR conditions (noise does not change speech boundaries).

3.3 Metrics

$$\text{F1} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}$$

$$\text{FAR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad \text{(false alarm rate on non-speech frames)}$$

$$\text{MISS} = \frac{\text{FN}}{\text{TP} + \text{FN}} \quad \text{(miss rate on speech frames)}$$

4 Results

4.1 Single-recording baseline (clean audio)

Table 1 shows performance on the unmodified `sample.wav`. The CNN VAD with threshold tuning achieves the highest F1 and dramatically lower false alarm rate compared to the energy baseline.

Table 1: Performance on clean `sample.wav`

Method	F1	FAR	MISS
Energy VAD (raw)	0.899	0.039	0.090
Energy VAD (smoothed)	0.922	0.022	0.090
CNN VAD (threshold = 0.5)	0.932	0.002	0.122
CNN VAD (threshold = 0.1)	**0.945**	0.015	**0.064**

Figure 1 shows the CNN's output probability trace over the recording. The model produces near-zero probabilities during silence and near-one probabilities during speech, with a clear bimodal distribution — confirming that the model has learned a well-separated decision boundary.

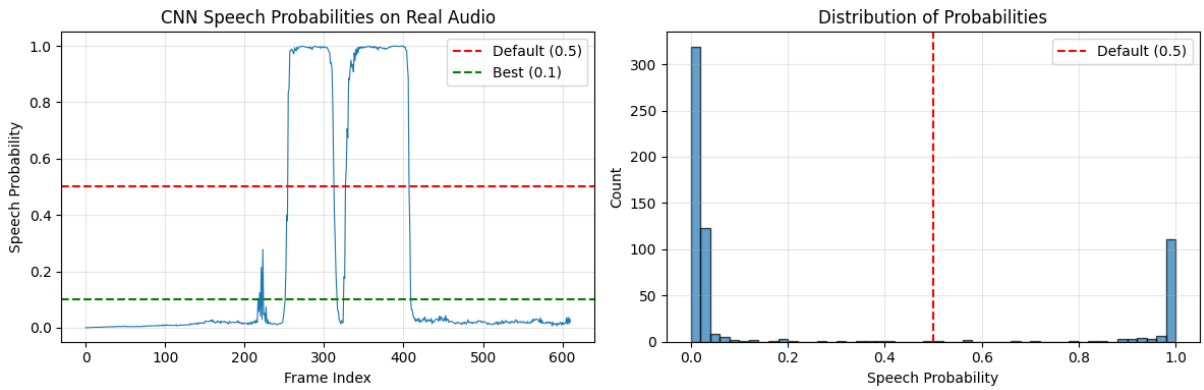

Figure 1: CNN speech probability on `sample.wav`. Left: probability trace over time (frame index). Right: histogram showing the bimodal distribution; most frames are confidently classified as silence or speech.

4.2 SNR stability sweep

Table 2 reports F1, FAR, and MISS for both methods at each SNR level. Figure 2 visualises the F1 and FAR curves. Note that 20 dB SNR is *not* equivalent to the clean recording in Table 1: noise is still present at 20 dB, so CNN F1 (0.922) remains slightly below the unmodified baseline (0.932).

Table 2: F1 and FAR at each SNR level (MUSAN noise mixing)

SNR (dB)	Energy F1	Energy FAR	CNN F1	CNN FAR
0	0.612	0.320	0.569	0.503
5	0.621	0.315	0.644	0.319
10	0.638	0.309	0.670	0.259
15	0.647	0.296	0.831	0.075
20	0.683	0.251	**0.922**	**0.007**

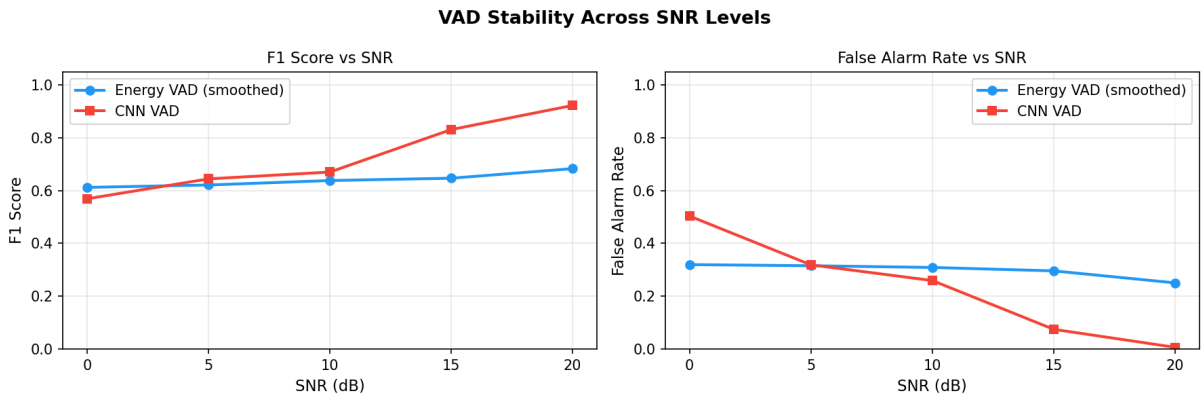

Figure 2: VAD stability across SNR levels. Left: F1 score vs. SNR. Right: False alarm rate vs. SNR. The crossover point on F1 occurs near 10 dB SNR.

4.3 Summary statistics

Table 3 aggregates the SNR sweep results. CNN VAD achieves a higher mean F1 but with 5× larger standard deviation, confirming the core finding.

Table 3: Aggregate statistics across SNR $\in \{0, 5, 10, 15, 20\}$ dB

	Energy VAD			CNN VAD		
Metric	Mean	Std	Min	Mean	Std	Min
F1	0.640	0.025	0.612	0.727	0.130	0.569
FAR	0.298	0.025	0.251	0.233	0.177	0.007
MISS	0.114	0.017	0.096	0.096	0.042	0.019

4.4 Inference latency

Both methods are measured over 50 runs on CPU (no GPU):

Table 4: Frame-level inference latency (CPU)

Method	Latency (ms/frame)	Std
Energy VAD	0.005	0.006
CNN VAD	0.340	0.133

CNN VAD is 71× slower per frame. At a 10 ms hop, Energy VAD consumes ≈0.05% of real-time budget per frame; CNN VAD consumes ≈3.4%. Both are viable for streaming, but CNN leaves substantially less headroom.

5 Analysis: Answering the Research Question

5.1 Is CNN VAD more stable?

No — in the low-SNR regime, Energy VAD is more stable.

The F1 standard deviation of Energy VAD across SNR levels is 0.025, compared to 0.130 for CNN VAD. At 0 dB SNR, CNN VAD underperforms Energy VAD (F1 0.569 vs. 0.612) and its FAR spikes to 0.503 — meaning half of all non-speech frames are misclassified as speech.

The model was trained with MUSAN noise mixed at 5–20 dB SNR. At 0 dB, the input distribution is out-of-training-range, which explains the collapse. Energy VAD, by contrast, uses an adaptive threshold (per-recording mean) that inherently adjusts to overall signal level — making it more resilient when noise is uniform and stationary.

5.2 Does CNN VAD offer better peak performance?

Yes — substantially, when SNR $\geq$ 10 dB.

At 15 dB SNR, CNN F1 is 0.831 vs. Energy 0.647 (+28%). At 20 dB, CNN achieves 0.922 vs. 0.683 (+35%), with FAR dropping to 0.007 compared to Energy's 0.251. The CNN's learned spectral features distinguish speech from stationary noise far more precisely than a simple energy threshold.

5.3 Crossover point

The two methods produce nearly identical F1 at ≈10 dB SNR (Energy 0.638, CNN 0.670). Below 10 dB, Energy VAD is the safer choice. Above 10 dB, CNN VAD is clearly superior.

6 Real-World Trade-off

6.1 Decision matrix

Table 5: Application-level recommendation

Scenario	Recommended
Edge / embedded device (no PyTorch, <1 MB)	Energy VAD
Uncontrolled noise (SNR < 10 dB)	Energy VAD
No labelled training data available	Energy VAD
Latency budget < 0.01 ms/frame	Energy VAD
Controlled environment (office, phone) SNR ≥ 10 dB	CNN VAD
Strict false alarm requirement	CNN VAD
System already running PyTorch inference	CNN VAD

6.2 Cascade design (future work)

A practical deployment strategy is to cascade the two detectors:

1. **Gate:** Energy VAD pre-screens every frame (0.005 ms overhead). Frames below the energy threshold are immediately rejected as silence.
2. **Refine:** CNN VAD runs only on frames that pass the energy gate. Since silence frames are filtered out, CNN processes only ∼20–30% of frames in a typical recording, reducing its effective CPU load by ∼70%.

This design inherits Energy VAD's robustness in truly silent regions and CNN VAD's discriminative power in noisy-but-present-speech segments. The expected result is: stable low FAR (from Energy gate) combined with high speech recall (from CNN refinement) — without the low-SNR fragility observed in standalone CNN VAD.

7 Conclusion

The experimental results partially confirm and partially refute the original hypothesis. CNN VAD is *not* uniformly more stable — it is fragile at low SNR due to training distribution mismatch. However, it is substantially more accurate at moderate-to-high SNR, with dramatically lower false alarm rates.

Energy VAD offers a reliable, deployable baseline with zero training cost and 71× lower latency. CNN VAD is the right choice when SNR is known to exceed ≈10 dB and the deployment environment supports PyTorch.

The most practically robust architecture would be a cascaded system combining both methods, leveraging the complementary strengths of each approach.